

Weekly Notice 2

Exercises, week 2. The first two exercises are based on Appendix E.

1. Let X_1 and X_2 be i.i.d. with $X_i \sim \text{Bin}(10, 1/2)$ for $i = 1, 2$.
Determine $\mathbb{E}[X_1 + X_2 | X_1]$ and $\mathbb{E}[X_1 \cdot X_2 | X_1]$.
2. Exercise 29(2).
3. Exercise 3. Try to solve (2) without using Theorem F.5(6).
4. Exercise 5.
5. Exercise 6(1).
6. Let $(\epsilon_n)_{n \geq 0}$ be i.i.d. with $\epsilon_n \sim N(0, 1)$ for all $n \in \mathbb{N}_0$.
 - (a) Is $(\epsilon_n)_{n \geq 0}$ a Gaussian process? Is it stationary?
 - (b) Let $X_n = \epsilon_0$ for all $n \in \mathbb{N}_0$. Is $(X_n)_{n \geq 0}$ a Gaussian process? Is it stationary?
 - (c) Are $(\epsilon_n)_{n \geq 0}$ and $(X_n)_{n \geq 0}$ versions?
 - (d) Is $(\epsilon_n + n)_{n \geq 0}$ a Gaussian process? Is it stationary?
7. Exercise 16.

Gaussian processes: It is perhaps worth noting that in discrete time, Gaussian stochastic processes can be defined as follows:

1. ($T = \mathbb{N}_0$) A real stochastic process $(Y_n)_{n \geq 0}$ is Gaussian if and only if $(Y_0, \dots, Y_n)^\top$ follows an $(n + 1)$ -variate normal distribution for every $n \in \mathbb{N}_0$.
2. ($T = \mathbb{Z}$) A real stochastic process $(Y_n)_{n \in \mathbb{Z}}$ is Gaussian if and only if $(Y_k, \dots, Y_{k+n})^\top$ follows an $(n + 1)$ -variate normal distribution for all $k \in \mathbb{Z}$ and $n \in \mathbb{N}_0$.